

CONTRIBUTIONS TO NORMAL SAMPLING THEORY

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1. A joint normal-Wishart distribution with variates dependent

This distribution, which is that of first and second degree sums of observations from a normal population with mean vector and covariance matrix that we shall without loss of generality take as $\underline{O}$ and $\underline{P}$ (the correlation matrix), is determined partly on account of itself (the population mean vector and covariance matrix may be known sufficiently as is the case in standardised educational tests) and partly for simplifying the derivations of distributions of certain conventional statistics. The present investigation forms part of a thesis for the Ph.D. degree.

KIBBLE (1941) considered a normal-gamma distribution derived as a series in terms of Laguerre polynomials from the characteristic function (cf) of a bivariate normal distribution, but there is no further connection between his work and mine.

Take for example the univariate $N(\underline{O}', \underline{P})$ population with variate x , $\underline{O}' = \underline{O}$ and $\underline{P} = 1$, and a random n -sample $\{x_j: j=1,2,\dots,n\}$ drawn from it. The particular normal-Wishart distribution (in terms of its density) that we derive, is $f(\underline{u}', \underline{t})$ where $\underline{u}' = \underline{p} = \sum_{j=1}^n x_j$ and $\underline{t} = \underline{a} = \sum x^2$. The cf $\phi(\underline{v}', \underline{\tau}, \underline{u}', \underline{t})$, where $\underline{v}'$ and $\underline{\tau}$ carry $\underline{u}'$ and $\underline{t}$, in this case is

$$\phi(a, \lambda; \underline{p}, a) = \int_{-\infty}^{\infty} \dots \int (2\pi)^{-1/2n} e^{-1/2 \sum x^2 + i a \sum x + i \lambda \sum x^2} \prod dx \text{ (with } \prod dx = \prod_{j=1}^n dx_j),$$

which upon 'removal' of $\sum$ is seen to be

$$= \left[\int_{-\infty}^{\infty} (2\pi)^{-1/2} e^{-1/2 x^2 + i a x + i \lambda x^2} dx \right]^n.$$

Completing the square with respect to x in the exponent of e , and then applying the normal integral, we see that the cf

$$= (1 - 2i\lambda)^{-1/2n} e^{-1/2na^2/(1-2i\lambda)}.$$

Fourier inversion of this, by completion of the square with respect to a , application of the normal integral and recognition of the remaining part of the cf as the cf of a chisquared variate, leads to

$$f(\underline{p}, a) = \frac{1}{\sqrt{2\pi n} 2^{1/2(n-1)} \Gamma(\frac{n-1}{2})} e^{-1/2a} (a - \underline{p}^2/n)^{1/2(n-3)}.$$

Since $da/dA = 1$ and $-\frac{1}{2}a = -p^2/2n - \frac{1}{2}A$ where $A = a - p^2/n$, we have

$$f(p, A) = \frac{1}{\sqrt{2\pi n}} e^{-p^2/2n} \cdot \frac{1}{2^{\frac{1}{2}(n-1)} \Gamma(\frac{n-1}{2})} e^{-\frac{1}{2}A} A^{\frac{1}{2}(n-3)}.$$

So p and A — or, what is essentially the same, $\bar{x}$ and s^2 — are independent normal and chisquared variates. a is also a chisquared variate (to see this, put $a = 0$ in the cf), but p and a are dependent; so $f(p, a)$ is a normal-chisquared distribution with its variates dependent. In the multivariate case we have a normal-Wishart distribution with $\underline{u}'$ and $\underline{t}$ dependent. (By a Wishart distribution we mean any distribution of the same form as the distribution of the covariance matrix.)

Had we tried to find the cf of p and A , we would have found difficulty in attempting to 'remove' the Σ since the exponent of e would now have contained Σ^2 . Further, it takes no more energy to derive $f(p, a)$ from its cf than to derive $f(p, A)$ from its cf; and $f(p, A)$ follows immediately from $f(p, a)$, and from it $f(A)$ and the independence of p and A . Similar ideas apply in the multivariate case. Moreover $f(\underline{u}', \underline{t})$ leads to the distribution of Hotelling's ratio.

Although p and a are statistically dependent, they are both uncorrelated and functionally independent in the x_j .

2. Derivation of the distribution

We do the derivation for the bivariate $N(\underline{Q}', \underline{P})$ with variates x and y , but we proceed in such a way that $f(\underline{u}', \underline{t})$ from the k -variate $N(\underline{Q}', \underline{P})$ can be written down immediately (and proved inductively if desired). The density (pdf) is again obtained via the cf, but streamlining devices are introduced e.g. to shorten the cumbersome task of completing squares.

We have $\underline{Q}' = (0, 0)$ and $\underline{P} = \begin{pmatrix} 1 & \rho_{12} \\ \rho_{12} & 1 \end{pmatrix}$, and we draw a random sample $\{(x_j, y_j); j=1, 2, \dots, n\}$. We derive $f(\underline{u}', \underline{t})$ where $\underline{u}' = (p, q)$ and $\underline{t} = \begin{pmatrix} a & d \\ d & b \end{pmatrix}$, while $p = \Sigma x$, $q = \Sigma y$, $a = \Sigma x^2$, $b = \Sigma y^2$ and $d = \Sigma xy$.

$$\phi(a, b, \lambda, \mu, \theta; p, q, a, b, d) = \left[\frac{1}{2\pi\sqrt{P}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^E dx dy \right]^n$$

$$\text{where } E = -\frac{1}{2P} (P_{11}x^2 + P_{22}y^2 + 2P_{12}xy) + i\lambda x^2$$

$$+ i\mu y^2 + i\theta xy + i\alpha x + i\beta y$$

$$= -\frac{1}{2P} (\lambda^* x^2 + \mu^* y^2 + 2\theta^* xy + 2a^* x + 2\beta^* y)$$

$$(\text{where } \lambda^* = P_{11} - 2P_{1\lambda}, a^* = -P_{1a}, \text{ etc.})$$

$$= -\frac{1}{2P} (\lambda^* x^2 + \mu^* y^2 + \nu^* z^2 + 2\theta^* xy + 2a^* xz + 2\beta^* yz - \nu^* z^2)$$

$$(\text{where } z = 1)$$

$$= -\frac{1}{2P} \left[(x, y, z) \begin{pmatrix} \lambda^* & \theta^* & a^* \\ \theta^* & \mu^* & \beta^* \\ a^* & \beta^* & \nu^* \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \nu^* z^2 \right]$$

$$= -\frac{1}{2P} \left[(x, y, z) \underline{MM}' \begin{pmatrix} x \\ y \\ z \end{pmatrix} - x' z^2 \right]$$

$$\text{where } \underline{M} = \begin{pmatrix} \frac{C_{\mu^* \nu^*}}{\sqrt{C_{\mu^* \nu^*}}} & 0 & 0 \\ \frac{C_{\mu^* \theta^*}}{\sqrt{C_{\mu^* \nu^*}}} & \frac{C_{\nu^*}}{\sqrt{C_{\mu^* \nu^*} C_{\nu^*}}} & 0 \\ \frac{C_{\theta^* \beta^*}}{\sqrt{C_{\mu^* \nu^*}}} & \frac{C_{\theta^*}}{\sqrt{C_{\mu^* \nu^*} C_{\nu^*}}} & \frac{C}{\sqrt{C_{\nu^*} C}} \end{pmatrix}, C = \begin{vmatrix} \lambda^* & \theta^* & a^* \\ \theta^* & \mu^* & \beta^* \\ a^* & \beta^* & \nu^* \end{vmatrix} \text{ and}$$

e.g. $C_{\mu^* \nu^*}$ = the cofactor of μ^* in the cofactor of ν^* in C)

$$= \frac{1}{2P} \left[\frac{(\sigma_{\mu^* x} + \sigma_{\theta^* y} + C_{\theta^* \beta^*} z)^2}{\sigma_{\mu^*}} + \frac{(\sigma y + C_{\theta^* \beta^*} z)^2}{\sigma_{\mu^* \sigma}} + \frac{(C z)^2}{\sqrt{\sigma C}} - x^2 z^2 \right]$$

$$(\text{where } \sigma = C_{\nu^*})$$

$$= -\frac{1}{2P} \left[\frac{u^2}{\sigma_{\mu^*}} + \frac{v^2}{\sigma_{\mu^* \sigma}} + \left(\frac{C}{\sigma} - \nu^* \right) \right] (\text{where } \mu = \sigma_{\mu^* x} + \sigma_{\mu^* y} + C_{\theta^* \beta^*} z$$

$v = \sigma y + C_{\theta^* \beta^*} z$; further $z = 1$ and ν^* is arbitrary);

$$\text{and } \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \sigma_{\mu}^* & \sigma_{\theta}^* \\ 0 & \sigma \end{vmatrix}^{-1} = (\sigma_{\mu}^* \sigma)^{-1},$$

so it follows by means of the normal integrals in u and v that

$$\begin{aligned} \phi(\underline{a}, \underline{\beta}, \lambda, \mu, \theta; p, q, a, b, d) &= \left[\frac{1}{2\pi\sqrt{P}} \sqrt{2\pi P \sigma_{\mu}^*} \sqrt{2\pi P \sigma_{\mu}^* \sigma} (\sigma_{\mu}^* \sigma)^{-1} \right]^n \\ &= P^{1/2n} \sigma^{-1/2n} e^{-n(C/\sigma - \nu^*)/2P} \\ &= P^{1/2n} \begin{vmatrix} \lambda^* & \theta^* \\ \theta^* & \mu^* \end{vmatrix}^{-1/2n} \exp\left(-n \begin{vmatrix} \lambda^* & \theta^* & a^* \\ \theta^* & \mu^* & \beta^* \\ a^* & \beta^* & 0^* \end{vmatrix} / 2P \begin{vmatrix} \lambda^* & \theta^* \\ \theta^* & \mu^* \end{vmatrix}\right) \text{ (which is the form of} \end{aligned}$$

the cf we will invert in order to get the pdf)

$$= P^{1/2n} \begin{vmatrix} \lambda^* & \theta^* \\ \theta^* & \mu^* \end{vmatrix}^{-1/2n} \exp \frac{n}{2P} (a^*, \beta^*) \begin{pmatrix} \lambda^* & \theta^* \\ \theta^* & \mu^* \end{pmatrix}^{-1} \begin{pmatrix} \nu^* \\ \beta^* \end{pmatrix}$$

(as we can see by expanding the 3×3 determinant according to its last row or column and then the cofactors according to their last columns or rows).

The above procedure allows straightforward stepwise and over-all generalisation of the result to

$$\phi(\underline{v}', \underline{x}; \underline{u}', \underline{t}) = P^{1/2n(k-1)} \Sigma^{-1/2n} e^{n\underline{v}'} \Sigma^{-1} \underline{v}/2P$$

where Σ is the k -dimensional equivalent of σ above and Σ the matrix with the same elements.

In the process of inversion we repeatedly employ the following integral:

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} z^{-n} e^{zk} dz = \frac{k^{n-1}}{\Gamma(n)} \quad (c \text{ and } c' \text{ real}). \dots (A)$$

If $c' = c > 0$, the integral follows from the Fourier transform of the gamma cf, and the general integral is an extension; but we find it more convenient than the Fourier transform.

$$f(p, q, a, b, d)$$

$$= \frac{P^{1/2}n}{(2\pi)^5} \int_{-\infty}^{\infty} \int \int \int \sigma^{-1/2} n \exp \left(-\frac{n}{2P\sigma} \begin{vmatrix} \lambda^* & \theta^* & a^* \\ \theta^* & \mu^* & \beta^* \\ a^* & \beta^* & 0^* \end{vmatrix} - i\alpha p - i\beta q - i\lambda a - i\mu b - i\theta d \right)$$

$$= \frac{P^{1/2}n}{(2\pi)^5} \int \int \int \int \sigma^{-1/2} n \exp \left(-\frac{n}{2P\sigma} \begin{vmatrix} \lambda^* & \theta^* & a^* \\ \theta^* & \mu^* & \beta^* \\ a^* & \beta^* & -\frac{2}{n}(a^*p + \beta^*q) \end{vmatrix} - i\lambda a - i\mu b - i\theta d \right).$$

$$da \, d\beta \, d\lambda \, d\mu \, d\theta$$

$$(\text{since } -i\alpha p - i\beta q = (a^*p + \beta^*q)/P)$$

$$= \frac{P^{1/2}n}{(2\pi)^5} \int \int \int \int \sigma^{-1/2} n \exp \left(-\frac{n}{2P\sigma} \begin{vmatrix} \lambda^* & \theta^* & \frac{a^* + \lambda^*p + \theta^*q}{n} \\ \theta^* & \mu^* & \frac{\beta^* + \theta^*p + \mu^*q}{n} \\ a^* + \frac{\lambda^*p + \theta^*q}{n} & \beta^* + \frac{\theta^*p + \mu^*q}{n} & \frac{\lambda^*p^2 + 2\theta^*pq + \mu^*q^2}{n} \end{vmatrix} \right.$$

$$\left. - i\lambda a - i\mu b - i\theta d \right) da \, d\beta \, d\lambda \, d\mu \, d\theta$$

(if we operate with the rows and columns of the third order determinant so as to get an element without a^* or β^* in position (3,3))

$$= \frac{P^{1/2}n}{(2\pi)^5} \int \int \int \int \sigma^{-1/2} n \exp \left(-\frac{nP}{2\sigma} \left(a + i \frac{\lambda^*p + \theta^*q}{nP}, \beta + i \frac{\theta^*p + \mu^*q}{nP} \right) \begin{pmatrix} \sigma_{\lambda^*} & \sigma_{\theta^*} \\ \sigma_{\theta^*} & \sigma_{\mu^*} \end{pmatrix} \right)$$

$$\left(a + i \frac{\lambda^*p + \theta^*q}{nP}, \beta + i \frac{\theta^*p + \mu^*q}{nP} \right) - \frac{1}{2nP} (\lambda^*p^2 + \mu^*q^2 + 2\theta^*pq)$$

$$- i\lambda a - i\mu b - i\theta d) da \, d\beta \, d\lambda \, d\mu \, d\theta$$

$$= \frac{P^{1/2}n}{(2\pi)^5} \int \int \int \int \sigma^{-1/2} n \exp \left(-\frac{nP}{2\sigma} \left[\left(\frac{\Delta \sigma_{\mu^*} (a + i \frac{\lambda^*p + \theta^*q}{nP}) + \Delta \sigma_{\theta^*} (\beta + i \frac{\theta^*p + \mu^*q}{nP})}{\sqrt{\Delta \sigma_{\mu^*}}} \right)^2 \right. \right.$$

$$\left. + \left(\frac{\Delta (\beta + i \frac{\theta^*p + \mu^*q}{nP})}{\sqrt{\Delta \sigma_{\mu^*} \Delta}} \right)^2 \right] - \frac{1}{2nP} (\lambda^*p^2 + \mu^*q^2 + 2\theta^*pq) -$$

$$- i\lambda a - i\mu b - i\theta d) da \, d\beta \, d\lambda \, d\mu \, d\theta$$

(where $\Delta = \begin{vmatrix} \sigma_{\lambda}^* \sigma_{\theta}^* \\ \sigma_{\theta}^* \sigma_{\mu}^* \end{vmatrix}$, and applying the normal integral and Jacobi's theorem by which $\Delta = \sigma^{k-1} = \sigma$ here, Δ being the conjugate of σ , we find that

$$\begin{aligned}
 f(p, q, a, b, d) &= \frac{p^{1/2}(n-2)}{n(2\pi)^4} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sigma^{-1/2}(n-1) \exp\left(-\frac{1}{2nP} (\lambda^* p^2 + \mu^* q^2 + 2\theta^* pq) - \right. \\
 &\quad \left. - i\lambda a - i\mu b - i\theta d\right) d\lambda d\mu d\theta \\
 &= \frac{p^{1/2}(n-2)}{n(2\pi)^4} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sigma^{-1/2}(n-1) \exp \frac{1}{2P} \left(\lambda^* \left(a - \frac{p^2}{n}\right) + \mu^* \left(b - \frac{q^2}{n}\right) \right. \\
 &\quad \left. + 2\theta^* \left(d - \frac{pq}{n}\right) \right) \exp\left(-\frac{1}{2P} (P_{11}a + P_{22}b + 2P_{12}d)\right) d\lambda d\mu d\theta \\
 &= \frac{p^{1/2}(n-2)}{n(2\pi)^4} \exp\left(-\frac{1}{2P} (P_{11}a + P_{22}b + 2P_{12}d)\right) \\
 &\quad \frac{1}{(2\pi i)^2 P_i} \int_{P_{12}-i\infty}^{P_{12}+i\infty} \int_{P_{22}-i\infty}^{P_{22}+i\infty} \int_{P_{11}-i\infty}^{P_{11}+i\infty} \sigma^{-1/2}(n-1) \\
 &\quad \exp \frac{1}{2P} (\lambda^* A + \mu^* B + 2\theta^* D) d\lambda^* d\mu^* d\theta^*
 \end{aligned}$$

(where $A = a - \frac{p^2}{n}$, etc.).

Now, the following cofactor expansion may be used:

$$\begin{aligned}
 \lambda^* A + \mu^* B + 2\theta^* D &= (T_B \sigma / T_{AB} \sigma_{\lambda}^* + T \sigma_{\lambda}^* / T_B \sigma_{\lambda^* \mu^*}) \\
 &\quad + (T_B (\theta^*)^2 / \sigma_{\lambda^*})
 \end{aligned}$$

$$\text{where } T = \begin{vmatrix} A & D \\ D & B \end{vmatrix}.$$

(The higher-order expansions are written down by strict adherence to the ascendance and descendance of the cofactors and working row by row in the case of the non-diagonal elements of σ , e.g. in the third order case we have

$$\begin{aligned}
 \lambda^* A + \mu^* B + \nu^* C + 2\theta^* D + 2\psi^* F + 2\phi^* E &= (T_{BC} \sigma / T_{ABC} \sigma_{\lambda^*} + T_c \sigma_{\lambda^*} / T_{BC} \sigma_{\lambda^* \mu^*} \\
 &\quad + T \sigma_{\lambda^* \mu^*} / T_C \sigma_{\lambda^* \mu^*} / T_C \sigma_{\lambda^* \mu^* \nu^*})
 \end{aligned}$$

$$+ (T_{BC} \sigma_{\lambda^* \mu^*} (\theta^{*'})^2 / T_{ABC} \sigma_{\lambda^*} + T_{BC} \sigma_{\lambda^*} + T_{BC} \sigma_{\lambda^* \mu^* \nu^*} (\psi^{*'})^2 / T_{ABC} \sigma_{\lambda^* \mu^*} \\ + T_{C} \sigma_{\lambda^* \mu^* \nu^*} (Q^{*'})^2 / T_{BC} \sigma_{\lambda^* \mu^*}) (\text{where}$$

$$T = \begin{vmatrix} A & D & F \\ D & B & E \\ F & E & C \end{vmatrix} \quad \text{and} \quad \sigma = \begin{vmatrix} \lambda^* \theta^* \psi^* \\ \theta^* \mu^* \varphi^* \\ \psi^* \varphi^* \nu^* \end{vmatrix};$$

further $\theta^{*'} = \theta + \text{term without it}$, and $\psi^{*'}$ and $\theta^{*''}$ similar).

Since $\theta^* = P_{12} - P_i \theta$ we have

$$f(p, q, a, b, d) = \frac{p^{1/2}(n-2)}{n(2\pi)^4} \exp \left(-\frac{1}{2P} (P_{11} a + P_{22} b + 2P_{12} d) \right)$$

$$\frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{P_{22}-i\infty}^{P_{22}+i\infty} \int_{P_{11}-i\infty}^{P_{11}+i\infty} \sigma^{-1/2(n-1)}$$

$$\exp \left(\frac{1}{2P} (T_B \sigma / T_{AB} \sigma_{\lambda^*} + T \sigma_{\lambda^*} / T_B \sigma_{\lambda^* \mu^*}) - \frac{P}{2} T_B (\theta^{*''})^2 / \right.$$

$$\sigma_{\lambda^*}) d\lambda^* d\mu^* d\theta^{*''} \quad (\theta^{*''} \text{ is like } \theta^{*'})$$

$$= \frac{p^{1/2}(n-2)}{n(2\pi)^4} \exp \left(-\frac{1}{2P} (P_{11} a + P_{22} b + 2P_{12} d) \right)$$

$$\frac{1}{(2\pi)^2} \int_{P_{22}-i\infty}^{P_{22}+i\infty} \int_{P_{11}-i\infty}^{P_{11}+i\infty} \sigma^{-1/2(n-1)} \exp \frac{1}{2P} (T_B \sigma / T_{AB} \sigma_{\lambda^*}$$

$$+ T \sigma_{\lambda^*} / T_B \sigma_{\lambda^* \mu^*}) (2\pi \sigma_{\lambda^*} / P T_B)^{1/2} d\lambda^* d\mu^* \quad (\text{using the normal integral})$$

$$= \frac{P^{1/2}(n-2)}{n(2\pi)^4} \exp \left(-\frac{1}{2P} (P_{11}a + P_{22}b + 2P_{12}d) \right)$$

$$\frac{1}{(2\pi i)^2} (2\pi/PT_B)^{1/2} \int_{-\infty-i\infty}^{+\infty+i\infty} \frac{P_{22} + i\infty}{P_{22} - i\infty} \sigma^{-1/2(n-1)}$$

$$\sigma_{\lambda^*}^{-1/2} \exp \frac{1}{2P} (T_B \sigma / T_{AB} \sigma_{\lambda^*} + T \sigma_{\lambda^*} / T_B \sigma_{\lambda^* \mu^*}) d\sigma d\sigma_{\lambda^*}$$

$$(\text{since } \partial(\lambda^*, \mu^*) / \partial(\sigma, \sigma_{\lambda^*}) = \sigma_{\lambda^*}^{-1})$$

$$= \frac{P^{1/2}(n-7)}{4n(2\pi)^{1/2} T_B^{1/2}} \exp \left(-\frac{1}{2P} (P_{11}a + P_{22}b + 2P_{12}d) \right)$$

$$\frac{1}{(2\pi i)^2} \int_{-\infty-i\infty}^{+\infty+i\infty} \frac{P_{22} + i\infty}{P_{22} - i\infty} \sigma^{-1/2(n-1)} \sigma_{\lambda^*}^{-1/2} \exp(T_B \sigma /$$

$$2PT_{AB} \bar{\sigma}_{\lambda^*}) \exp (T \sigma_{\lambda^*} / 2PT_B \sigma_{\lambda^* \mu^*}) d\sigma d\sigma_{\lambda^*}$$

$$= \frac{T_B^{1/2}(n-4)}{n(2\pi)^{1/2} \Gamma(\frac{n-1}{2}) 2^{1/2}(n+1)} \exp \left(-\frac{1}{2P} (P_{11}a + P_{22}b$$

$$+ 2P_{12}d) \right) \frac{1}{2\pi i} \int_{P_{22} - i\infty}^{P_{22} + i\infty} \sigma_{\lambda^*}^{-1/2(n-2)} \exp (T \sigma_{\lambda^*} /$$

$$2PT_B \sigma_{\lambda^* \mu^*}) d\sigma_{\lambda^*}.$$

$$= \frac{1}{n\pi^{1/2} \Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) 2^n P^{1/2n}} \exp \left(-\frac{1}{2P} (P_{11}a$$

$$+ P_{22}b + 2P_{12}d) \right) T^{1/2(n-4)}$$

(where we have applied the integral (A)).

Again straightforward stepwise and over-all generalisation can take place:

$$\begin{aligned}
 f(\underline{u}', \underline{t}) &= [(2\pi n)^{1/2k} \sqrt{P} \cdot 2^{1/2k(n-1)} \pi^{-1/4k(k-1)} \prod_{i=1}^k \Gamma(\frac{n-i}{2}) P^{1/2(n-1)}]^{-1} \\
 &\quad \exp(-1/2 \operatorname{tr} \underline{P}^{-1} \underline{t}) T^{1/2(n-k-2)}; \\
 \text{or } (2\pi n)^{-1/2k} P^{-1/2} \exp(-1/2 \underline{u}' \underline{P}^{-1} \underline{u}) \\
 &\quad 2^{-1/2k(n-1)} \pi^{-1/4k(k-1)} \prod_{i=1}^k \Gamma(\frac{n-i}{2}) P^{-1/2(n-1)} \\
 &\quad \exp(-1/2 \operatorname{tr} \underline{P}^{-1} \underline{T}) T^{1/2(n-k-2)}.
 \end{aligned}$$

3. Deductions from the distribution

3.1 *The Wishart distribution and the independence of mean vector and covariance matrix:*

Since $\partial(a, b, \dots, d, \dots) / \partial A, B, \dots, D, \dots = 1$ the expression for $f(\underline{u}', \underline{T})$ is the same as the last one for $f(\underline{u}', \underline{t})$.

$f(\underline{u}', \underline{T})$, then, factor as above, and since the factor in $\underline{u}'$ is $f(\underline{u}')$, the other is the Wishart distribution $f(\underline{T})$ of $\underline{T}$, and moreover $\underline{u}'$ and $\underline{T}$ are independent.

Alternatively, $\phi(\underline{\tau}; \underline{t})$ may be found (from the second last paragraph of section 1 we know this to be slightly easier than to find $\phi(\underline{\tau}; \underline{T})$, and from it $f(\underline{t})$ (same effort as $f(\underline{T})$ from $\phi(\underline{\tau}; \underline{T})$), then $f(\underline{T})$ follows from $f(\underline{t})$ by decreasing the number of degrees of freedom by 1 (i.e. writing $n-1$ for n) and replacing $\underline{t}$ and its elements $a, b, \dots, d, \dots$ by $\underline{T}$ and its corresponding elements $A, B, \dots, D, \dots$

$$\text{When } k=1, 2 \quad \underline{t} = \begin{pmatrix} a & d \\ d & b \end{pmatrix}, \quad \begin{pmatrix} a & d & f \\ d & b & e \\ f & e & c \end{pmatrix}, \quad \text{and}$$

$$\begin{aligned}
 f(a, b, d) &= \frac{1}{2^n \sqrt{\pi} \Gamma(\frac{n}{2}) \Gamma(\frac{n-1}{2}) P^{1/2n}} e^{-\frac{1}{2P} (a - 2\zeta d + b)} (ab - d^2)^{1/2(n-3)} \\
 &\quad (n > 3)
 \end{aligned}$$

$$f(a,b,c,d,e,f) = \frac{1}{2^{1/2n} \pi^{1/2} \Gamma(\frac{n}{2}) \Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) P^{1/2n}}$$

$$\exp \left(-\frac{1}{2P} (P_{11}a + P_{22}b + P_{33}c + 2P_{12}d + 2P_{23}e + 2P_{31}f) \right)$$

$$(abc + 2def - ae^2 - bf^2 - cd^2)^{1/2(n-4)} \quad (n > 4)$$

$$a, b, c \geq 0, |d| \leq \sqrt{ab}, |e| \leq \sqrt{bc} \text{ and } |f| \leq \sqrt{ca}.$$

3.2 The distribution of the generalised Student's ratio of Hotelling:

Let V be the generalised variance and v the corresponding statistic but about the population mean vector instead of the sample mean vector, then we are to find the distribution of $V/v = n^k T/n^k t = T/t$ (where T and t are of order k). We find the distribution for $k = 3$, but again in such a way that the distribution for $k = 1, 2, 4, 5, 6, \dots$ can immediately be written down. ($u' = (p, q, r) = (\Sigma x, \Sigma y, \Sigma z)$, z like x, y .)

$$T = \begin{vmatrix} a - p^2/n & d - pq/n & f - pr/n \\ d - qp/n & b - q^2/n & e - qr/n \\ f - rp/n & e - rq/n & c - r^2/n \end{vmatrix} = \begin{vmatrix} a & d & f \\ d & b & e \\ f & e & c \end{vmatrix} - \frac{1}{n} \left(p \begin{vmatrix} d & f \\ q & e \\ r & c \end{vmatrix} + q \begin{vmatrix} a & f \\ d & e \\ f & c \end{vmatrix} + r \begin{vmatrix} a & d \\ d & b \\ f & e \end{vmatrix} \right)$$

$$= t - \frac{1}{n} \{ p(pt_a + qt_d + rt_f) + q(pt_d + qt_b + rt_e) + r(pt_f + qt_e + rt_c) \}$$

$$= t - \frac{1}{n} (p^2 t_a + q^2 t_b + r^2 t_c + 2pqt_d + 2qrt_e + 2rpt_f)$$

$$= t \left(1 - \frac{\begin{vmatrix} p & d & f \\ q & b & e \\ r & e & c \end{vmatrix}^2}{ntt_a} - \frac{\begin{vmatrix} q & e \\ r & c \end{vmatrix}^2}{nt_a t_{ab}} - \frac{|r|^2}{nt_{ab} t_{abc}} \right)$$

(and the squares here are similar to those obtained in the expression for E in section 2, only they are written in terms of Kramer determinants)

$$= t(1 - a - \beta - \gamma) \text{ say,}$$

and $T/t = 1 - a - \beta - \gamma$.

Now, $f(p, q, r, t) = K e^{-1/2 \text{tr} P^{-1} t} T^{1/2(n-5)}$ (K is the norming constant of $f(\underline{u}', t)$)

$$= K e^{-1/2 \text{tr} P^{-1} t} t^{1/2(n-5)} (1 - a - \beta - \gamma)^{1/2(n-5)}$$

$$\text{Further, } \frac{\partial(p,q,r)}{\partial(a,\beta,\lambda)} = \begin{vmatrix} 2\sqrt{t_a}/nta^{1/2} & ? & ? \\ 0 & 2\sqrt{t_{ab}}/nt_a\beta^{1/2} & ? \\ 0 & 0 & 2\sqrt{t_{abc}}/nt_{ab}\lambda^{1/2} \end{vmatrix}^{-1}$$

(the question marks indicate that the particular elements are unimportant).

$$\text{So } f(a,\beta,\gamma,t) = n^{-1/2} K e^{-1/2 \text{tr} P^{-1} t} t^{1/2(n-4)} (1-a-\beta-\gamma)^{1/2(n-5)} (a\beta\gamma)^{-1/2}$$

(where $0 \leq a, \beta, \gamma \leq 1$),

and it follows (using the facts that the sum of Dirichlet variates is a beta variate and that if δ is a beta variate then $1-\delta$ is another, as well as the knowledge of what happens to the various parameters) that T/t is independent of t , that T/t is $\text{Be}((n-3)/2, 3/2)$ or $\text{B}((n-k)/2, k/2)$ in the k -variate instance, and also that $f(t)$ has the same distribution as found for $f(T)$ above but with t and its elements instead of T and its corresponding elements and with the number of degrees of freedom increased by 1 (i.e. with n instead of $n-1$).

$$\begin{aligned} \text{Since } a+\beta+\gamma &= \frac{1}{nt} (p^2 t_a + q^2 t_b + r^2 t_c + 2pqt_d + 2qrt_e + 2rpt_f) \\ &= \frac{1}{nt} (p,q,r) \begin{pmatrix} t_a & t_d & t_f \\ t_d & t_b & t_e \\ t_f & t_e & t_c \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} \\ &= \frac{1}{n} (p,q,r) \begin{pmatrix} a & d & f \\ d & b & e \\ f & e & c \end{pmatrix}^{-1} \begin{pmatrix} p \\ q \\ r \end{pmatrix}, \end{aligned}$$

we have, if we generalise, that $\frac{1}{n} \underline{u}' t^{-1} \underline{u}$ is a $\text{Be}(\frac{k}{2}, \frac{n-k}{2})$ variate, while $\frac{1}{n} \underline{u}' P^{-1} \underline{u}$ is a $G(\frac{k}{2})$ variate using WILKS's (1962) notations).

In the next section we consider further deductions from $f(\underline{u}', t)$, as well as deductions from $f(\underline{u}')$ and $f(t)$.

4. Various derived distributions, regressions and scedasticities

$$4.1.1 \quad f(t|\underline{u}') = f(\underline{t}, \underline{u}')/f(\underline{u}') = f(T)f(\underline{u}')/f(\underline{u}') = f(T),$$

but e.g. $f(b,p) \neq f(b-p^2/n)$ since $\phi(a,\beta,\lambda,\mu,\theta;p,q,a,b,d)/\beta=\lambda=\theta=0, \mu=\lambda \neq \phi(a,\beta,\lambda,u,\theta;p,q,a,b,d)/\beta=\mu=\theta=0$.

$$4.1.2 \quad f(\underline{u}'|t) = (f(T)/f(t))f(\underline{u}')$$

and $\underline{u}'$ and $\underline{T}$ are independent, but to say

$$E(u'|t) = (f(T)/f(t))E(u')$$

does not make sense. However,

$$E(p|a) = \int_{-\infty}^{\infty} \int_{p^2/n}^{\infty} p f(p,a)/f(a) da dp = 0$$

since the integrand with respect to p is odd in p . Further $f(p|a,b,d) = K(1-p^2/na)^{1/2(n-5)}$ where K does not involve p , and $E(p|a,b,d) = 0$ (the integration limits are $\pm\infty$); similarly $E(q|a,b,d) = 0$; thus $E(p,q|a,b,d) = (0,0)$. In general, $E(\underline{u}'|t) = E(\underline{u}')$.

4.2 We now show how conditionality may be used to derive certain p.d.f.'s regressions and scedasticities from others.

4.2.1 From the expression for $f(a,b,d)$ we have that

$$f(t,b,d) = \frac{1}{2^n \sqrt{\pi} \Gamma(\frac{n}{2}) \Gamma(\frac{n-1}{2}) P^{1/2} n} \exp(-\frac{t}{2Pb} - \frac{b}{2} - (\frac{d}{2Pb} \rho b)^2) t^{1/2(n-3)} b^{-1/2}$$

since $t=ab-d^2$, $1-\rho^2=P$ and $da/dt = (dt/da)^{-1} = b^{-1}$;

then integrating out d as a conditional $N(\rho b, Pb)$ variate (we also note that $d|(t,b) = d|b$, we have that

$$f(t,b) = \frac{1}{2^{1/2}(2n-1) \Gamma(\frac{n}{2}) \Gamma(\frac{n-1}{2}) P^{1/2} (n-1)} \exp(-\frac{t}{2Pb} - \frac{b}{2}) b^{-1/2} t^{1/2(n-3)};$$

and, integrating out t as a conditional gamma variate, we have that

$$f(b) = \frac{1}{2^{1/2} n \Gamma(\frac{n}{2})} e^{-1/2 b} b^{1/2(n-2)};$$

$$\text{So } f(t|b) = \frac{1}{2^{1/2}(n-1) \Gamma(\frac{n-1}{2}) P^{1/2} (n-1)} e^{-t/2Pb} t^{1/2(n-3)} b^{-1/2(n-1)}.$$

Now, consider T of first order, A say; and, integrating out t as a conditional gamma variate, we have that $f(b) = 1/2^{1/2} n \Gamma(n/2) e^{-1/2 b} b^{1/2(n-2)}$; by putting pivotal condensation in reverse we see that $T = t/ny^2$ where t is of second order and y constant. A constant is a *variable* with single point domain, so we may consider $\text{cov}(x,y)$, and we find it is $E(xy) = yE(x) = 0$; further, the important part of a correlation coefficient is the covariance (the variances merely serve a norming purpose), and so $\rho(x,y) = 0$ iff $\text{cov}(x,y) = 0$, as is the position here. $P=1-\rho^2 = 1$. From the expression for $f(t|b)$ above therefore follows that

$$f(ny^2 A) = \frac{1}{2^{1/2}(n-1) \Gamma(\frac{n-1}{2})} e^{-1/2 A} A^{1/2(n-3)} (ny^2)^{-1}$$

$$\text{since } b = \Sigma y^2 = ny^2; \text{ and } f(A) = \frac{1}{2^{1/2}(n-1) \Gamma(\frac{n-1}{2})} e^{-1/2 A} A^{1/2(n-3)}.$$

$$4.2.2 \quad f(a|b,d) = f(a,b,d)/f(b,d) = \frac{1}{(2P)^{1/2}(n-1)\Gamma(\frac{n-1}{2})} e^{-(\frac{t}{b})/2P} \left(\frac{t}{b}\right)^{\frac{n-3}{2}},$$

and if, as in 4.2.1, we let y be a constant, so that $d = yp$, $b = ny^2$, $t = ny^2A$ and $\rho = 0$, we have

$$f(a|p) = f(A);$$

and as regards regression and scedasticity we have

$$\text{regr}(a|b,d) = P(n-1) + d^2/b \text{ and } \text{sced}(a|b,d) = 2(n-1)P^2,$$

and

$$\text{and } \text{regr}(a|p) = (n-1) + p^2/n \text{ and } \text{sced}(a|p) = 2(n-1).$$

Similarly $f(d|b) = n(d, \rho b, Pb)$ leads to $n(p, 0, n)$ and $f(a, d|b)$ to $f(a, p)$.

$$4.2.3 \quad f(b,d) = \int_0^\infty f(a,b,d) da = \frac{1}{2^{1/2}(n+1)\sqrt{\pi} \Gamma(\frac{n}{2}) \sqrt{P}} \exp\left(-\frac{1}{2Pb}(d^2 - 2\rho db + b^2)\right) b^{1/2}(n-3)$$

while

$$f(p,q) = \frac{1}{2n\pi\sqrt{P}} \exp\left(-\frac{1}{2nP}(p^2 - 2\rho pq + q^2)\right),$$

but the strong agreement is not explained by conditionality: Take $n=1$, then $f(b,d) = f(y^2, xy)$ and $f(p,q) = f(x,y)$, but $\partial(y^2, xy)/\partial(x,y) = 2y^2$ and y^2 has half the domain of y . This simple link serves at least as a check for the correctness of the one p.d.f. given the other.

4.3 Further interesting p.d.f.'s (etc.) are the following

4.3.1 $f(a|b,d)$ and (as may be shown directly) $f(\frac{t}{b}|b,d)$ are equal to the same expression viz. that given in 4.2.2.

4.3.2 $f(\frac{t}{b}|b,d) = f(\frac{t}{b})f(b,d)$ (as follows directly from $f(a,b,d)$), so $f(\frac{t}{b}|b,d) = f(\frac{t}{b})$, which means that giving (b,d) leaves the same information in t that dividing by b does. As may now be expected, $f(t|b) = f(t|b,d)$, which means that t as reduced of information by b , is stable as regards d (i.e. $t|b$ is independent of d : $(t|b)|d = t|b$).

4.3.3 $f((a,d)|t) = f(a,t,d)/f(t) = f(a,b,d)/af(t) = (2\pi a)^{-1/2} \exp(-(\frac{t}{b})^2 - a)^2/2a) \cdot (2\pi Pa) \exp(-(d-\rho a)^2/2Pa) = f(a|t)f(d|a)$. But $f((a,d)|t) = f(a|t,d|t) = f(a|t)f((d|t)|(a|t))$, so $f((d|t)|(a|t)) = f(d|a)$, a fact which is not obvious. (Note that $f(a|t)$ looks like $n(\frac{t}{b})^{1/2}; a, a$) but is in fact what we shall presently call a kappa p.d.f., while $d|b$ which is a kappa function variate is a conditional $N(\rho b, P b)$ variate.)

4.4 *The nondiagonal element $d = \Sigma xy$*

$$4.4.1 \quad f(d) = \int_0^\infty f(b,d) db = \frac{2}{(2\pi)^{1/2} 2^{1/2n} \Gamma(\frac{n}{2}) \sqrt{P}} (\sqrt{d^2})^{1/2(n-1)} K_{1/2(n-1)} (\sqrt{\frac{d^2}{P^2}})$$

(or, if we take into account P_{11} and P_{22} which here are both 1, we have

$$f(d) = \frac{2}{(2\pi)^{1/2} 2^{1/2n} \Gamma(\frac{n}{2}) (P_{11} P_{22})^{1/4(n-1)} (\sqrt{d^2})^{1/2(n-1)} K_{1/2(n-1)} \sqrt{\frac{P_{11} P_{22} d^2}{P^2}}}$$

We pay special attention to the following integral applied here:

$$\int_0^\infty e^{-a/x - bx} x^{m-1} dx = 2(a/b)^{1/2m} K_m(2\sqrt{ab}), \quad a, b > 0;$$

see T.I.T.1,4.5(29).* We refer to it as the kappa integral because of the kappa modified Bessel function involved. By a kappa p.d.f. we mean one of the type

$$f(x) = e^{-a/x - bx} x^{m-1} / 2(a/b)^{1/2m} K_m(2\sqrt{ab})$$

(compare this with the gamma p.d.f. where the norming constant contains a gamma function, the beta p.d.f. where it contains a beta function, etc.) By a kappa function p.d.f. we mean one whose kappa function contains the variate. The class of kappa function distributions include amongst others the gamma and the normal distributions since $e^{-z} = (2z/\pi)^{1/2} K_{1/2}(z)$.

4.4.2 If $n=1$ $d=xy$, so the product of two correlated identically distributed normal variates is distributed as xy in

T.I.T. and H.T.F.: See Erdélyi et al., literature list.

$$f(xy) = \frac{1}{\pi\sqrt{P}} K_0(|xy|/P) e^{\rho xy/P} \text{ or } \frac{1}{\pi} K_0(|xy|) \text{ if } \rho(x,y) = 0.$$

This result is of course not new (this derivation is); later we show how the p.d.f. of a product of several such normal variates is derived.

4.4.3 For even n , $f(d)$ is a linear sum of Laplace distributions. Applying $K_{r+1}(w) = K_{r-1}(w) + \frac{2r}{w} K_r(w)$, $K_{1/2}(w) = (\pi/2w)^{1/2} e^{-w}$ and $K_{-r}(w) = K_r(w)$, we have for e.g. $n=2,4,6$ that

$$f(d) = \frac{1}{2}E, \frac{1}{4}(P+|d|)E, \frac{1}{16}(3P^2 + 3P|d| + d^2)E$$

where $E = e^{-|d|/P+d/P}$. Note that $|w| = \sqrt{w^2}$, and that $\frac{1}{2}, \frac{1}{4}, \frac{1}{16}$ are the values of $1/2^{1/2n} \Gamma(\frac{1}{2}n)$, which is the norming constant for $f(a)$ and $f(b)$ as well, when $n=2,4,6$.

4.4.4 Since x is an $N(0,1)$ variate, $x^2 = a$ has distribution

$$f(a) = \frac{1}{2^{1/2n} \Gamma(\frac{n}{2})} e^{-1/2a} a^{1/2(n-2)}.$$

When $\rho(x,y)=1$, we may consider $y=x$, so

$$\lim_{\rho \rightarrow 1} f(d) = f(a).$$

Setting $d=a=1$ say (choosing another positive value does not affect the issue much, but we shall not pursue the matter further) and $P = 1/z$, we find

$$\lim_{z \rightarrow \infty} K_{1/2(n-1)}(z) / \frac{\Gamma(\frac{n-1}{2})}{\Gamma(\frac{n}{2})} \sqrt{\pi/2z} e^{-1/2-(z^2-z)^{1/2}} = 1,$$

which has as special case the well-known asymptotic expression

$$\lim_{z \rightarrow \infty} K_m(z)/K_{1/2}(z) = 1.$$

4.4.5 To prove that $D = \sum_{j=1}^n (x_j - \bar{x})(y_j - \bar{y})$ is distributed as $d = \sum_{j=1}^n x_j y_j$, but with $n-1$ instead of n degrees of freedom, we use the following simple extension of the technique used correspondingly for A and a say: We make an orthogonal transformation from (x,y) to (ξ, η) , putting $\xi_1 = \sqrt{n} \bar{x}$ and $\eta_1 = \sqrt{n} \bar{y}$ and letting the coefficients of the other ξ be the same as the corresponding coefficients in the η_j ; then $\sum_{j=1}^n \xi_j^2 = \sum_{j=1}^n \eta_j^2$

$\sum_{j=1}^n \eta_j^2 = \sum_{j=1}^n y_j^2$, $\sum_{j=1}^n \xi_j \eta_j = \sum_{j=1}^n x_j y_j$ and $\partial(x_1, \dots, x_n, y_1, \dots, y_n) / \partial(\xi_1, \dots, \xi_n, \eta_1, \dots, \eta_n) = 1$,
and so the $(\xi_j \eta_j)$

are distributed independently (use the bivariate p.d.f.) and as the (x_j, y_j) . Since $A = \sum_{j=2}^n \xi_j^2$, $B = \sum_{j=2}^n \eta_j^2$ and $D = \sum_{j=2}^n \xi_j \eta_j$, this means that A,B,D are distributed as a,b,d but with n-1 replacing n.

4.5 The diagonal of $t(2 \times 2)$

4.5.1 The joint distribution of $g = \frac{1}{2}(a-b)$ and d may be obtained from $f(a,b,d)$ by setting $h = \frac{1}{2}(a+b)$, thereby obtaining $f(g,h,d)$, and then integrating out h by T.I.T.1, 4.3(11). We find

$$f(g,d) = \frac{2P^{\frac{1}{2}n-2}}{2\pi 2^{\frac{1}{2}n} \Gamma(\frac{1}{2}n)} e^{\rho d/P} (\sqrt{(g^2+d^2)/P^2})^{\frac{1}{2}(n-2)} K_{\frac{1}{2}(n-2)} (\sqrt{\frac{g^2+d^2}{P^2}}),$$

which reduces to

$$f(g,d) = \frac{2}{2\pi 2^{\frac{1}{2}n} \Gamma(\frac{1}{2}n)} (\sqrt{g^2+d^2})^{\frac{1}{2}(n-2)} K_{\frac{1}{2}(n-2)} (\sqrt{g^2+d^2})$$

if $\rho = 0$.

From symmetry we have that for $\rho=0$ g and d have the same p.d.f. viz. that of w say in

$$f(w) = \frac{2}{(2\pi)^{\frac{1}{2}} 2^{\frac{1}{2}n} \Gamma(\frac{1}{2}n)} (\sqrt{w^2})^{\frac{1}{2}(n-1)} K_{\frac{1}{2}(n-1)} (\sqrt{w^2})$$

as follows from 4.4.1 of this section.

(The fact that for $\rho=0$ g and d have the same p.d.f. is also obvious from the fact that $(x+y)\sqrt{2}$ and $(x-y)\sqrt{2}$ are $N(0,1)$ variates with correlation ρ then since $E(x+y)(x-y) = E(x^2 - y^2) = 0$).

If ρ is not necessarily zero,

$$f(g) = \frac{2}{(2\pi)^{\frac{1}{2}} 2^{\frac{1}{2}n} \Gamma(\frac{1}{2}n) \sqrt{P}} (\sqrt{g^2/P})^{\frac{1}{2}(n-1)} K_{\frac{1}{2}(n-1)} (\sqrt{g^2/P}).$$

g and d are not independent, even when $\rho=0$; however, they are uncorrelated, and they are functionally independent with respect to the observations x_j and y_j .

$$4.5.2 \ f(a,b) = \frac{1}{2^n \Gamma^2(\frac{1}{2}n) P^{\frac{1}{2}n}} e^{-(a+b)/2P} (ab)^{\frac{1}{2}(n-2)} {}_0F_1(\frac{1}{2}n; [\frac{\rho}{2P}]^2 ab)$$

(We assume that this is known; one may obtain it from $f(a,b,d)$ by transforming from d to $d/\sqrt{ab}$, applying T.I.T.1.3.2(13), noting that $d/P = -i.i$ d/P , and using ${}_1F_1(m; 2m; 2w) = e^w {}_0F_1(m + \frac{1}{2}; w^2/4)$ and the duplication formula for the gamma function.)

If $\rho=0$ $f(a,b) = f(a)f(b)$ i.e. a and b are independent then; this is to be expected since the bivariate normal p.d.f. is symmetric in x and y , and a and b are the same function of the x_j and y_j respectively. However, $f(a,b)$ is not bivariate reproductive in the degrees of freedom, unless a and b are uncorrelated: If (a_1, b_1) is independent of (a_2, b_2) , and (a_s, b_s) their sum, we find by writing the ${}_0F_1$ in their series forms and integrating term by term (a_2/a_s and b_2/b_s are beta variates) that

$$f(a_s, b_s) = \frac{1}{2^{2n} \Gamma^4(\frac{1}{2}n) P^n} e^{-(a_s + b_s)/2P} \sum_{r,s=0}^{\infty} \left\{ \frac{\Gamma^2(\frac{1}{2}n)(\rho/2P)^{2r+2s}}{\Gamma(\frac{1}{2}n+r) \Gamma(\frac{1}{2}n+s) r!} \right. \\ \left. \frac{(a_s b_s)^{r+s+n-1}}{s!} B^2(\frac{1}{2}n+s, \frac{1}{2}n+r) \right\}$$

which reduces to

$$f(a_s, b_s) = \frac{1}{2^{2n} \Gamma^2(n) P^n} e^{-(a_s + b_s)/2P} (a_s b_s)^{n-1} {}_0F_1(n; [\frac{\rho}{2P}]^2 a_s b_s) \text{ iff } \rho=0.$$

4.6 The non-diagonal elements and further kappa function distributions

$$4.6.1 \ \frac{1}{(2\pi)^k} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(1 + \sum_{j=1}^k a_j^2\right)^{-\frac{1}{2}n} \exp\left(-i \sum_{j=1}^k a_j x_j \prod_{j=1}^k da_j\right) = \\ \frac{2}{(2\pi)^{\frac{1}{2}k} 2^{\frac{1}{2}n} \Gamma(\frac{1}{2}n)} \left(\sqrt{\sum_{j=1}^k x_j^2}\right)^{\frac{1}{2}(n-k)} K_{\frac{1}{2}(n-k)}\left(\sqrt{\sum_{j=1}^k x_j^2}\right) \text{ for all natural } k.$$

To derive the case $k+1$ from the case k , take a_{k+1}^2 with the 1, transform from the a_j to $a_j/(1+a_{k+1}^2)$ and use T.I.T.1.1.13(45); to prove the case $k=1$, use T.I.T.1.1.3(7).

4.6.2 To find $f(d,f)$, where $d=\Sigma xy$ and $f=\Sigma xz$, we first find the characteristic function of d and f by setting in the c.f. of the trivariate normal-Wishart distribution (section 2 end) all carriers zero excepting those of d and f . It is of the form (say)

$$\phi(\theta, \varphi; d, f) = g^{-\frac{1}{2}n} (1 + h^2 [\theta + j\varphi + \ell]^2 + m^2 [\varphi + s]^2)^{-\frac{1}{2}n}.$$

Put $h(\theta + j\varphi + \ell) = a_1$ and $m(\varphi + s) = a_2$, and make the necessary change in the kernel of the Fourier exponential transform. Then we obtain by the result in 4.6.1 that

$$f(d, f) = \frac{2}{2\pi^{1/2} n \Gamma(1/2 n) P_{11}^{1/4} (n-2) \sqrt{P}} \exp\left(-\frac{1}{2P} (2P_{12}d + 2P_{13}f)\right) \\ (P_{22}d^2 + 2P_{23}df + P_{33}f^2)^{(n-2)/4} K_{1/2(n-2)}((P_{11}(P_{22}d^2 + 2P_{23}df + P_{33}f^2)/P^2)^{1/2}).$$

We may generalise to the p.d.f. of the non-diagonal elements of any row of the k-variate covariance matrix.

4.6.3 $f(d, e, f)$ offered difficulty (where $e = \Sigma yz$): If we proceed from $f(a, b, c, d, e, f)$, integrate out $a - (bf^2 - 2def + cd^2)/(bc - e^2)$ after having transformed from a to it (as a gamma variate), and then integrate out c (as a conditional kappa variate), we find that

$$f(u, d, e, f) = K \exp\left(-\frac{1}{2P} (2P_{12}d + 2P_{23}e + 2P_{31}f)\right) \Gamma\left(\frac{n-2}{2}\right) (2P/P_{11})^{1/2(n-2)} \\ 2 K_0\left(\frac{1}{P_u} ((P_{11}u + P_{11}e^2)(P_{22}u + P_{11}f^2)(P_{33}u + P_{11}d^2)/P_{11})^{1/2}\right) \\ \exp(2P_{11}def/2Pu) u^{1/2(n-4)}$$

(we have put $bc - e^2 = u$), and $f(d, e, f)$ is the integral of this with respect to u between the limits 0 and ∞ . When $n=1$, this complicated integral is determined indirectly: $d=xy$, $e=yz$ and $f=zx$, $f(x, y, z)$ is known and $\text{mod}(\partial(x, y, z))/(\partial(d, e, f)) = 1/2 |xyz|^{-1} = 1/2 |def|^{-1/2}$, and it follows that

$$f(d, e, f) = \frac{1}{2(2\pi)^{3/2} \sqrt{P}} \exp\left(-\frac{1}{2P} (P_{11}fd/e + P_{22}de/f + P_{33}ef/d + 2P_{12}d + 2P_{23}e + 2P_{31}f)\right) |def|^{-1/2}.$$

5. The distribution of the variance determinant V

The Mellin transform $g(s)$ of a p.d.f. $f(x)$ ($=0$ for $x < 0$) is the $(s-1)$ st origin moment thereof. The inverse Mellin transform then is

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} g(s) x^{-s} ds,$$

where the purpose of the arbitrary real constant c is that the path of integration will not

pass through the pole (if any) at the origin. The so-called G functions of C.S. Meijer form an important subclass of the inverse Mellin transforms.

From the expression for $f(T)$ (section 3.1) we find, by means of the norming constant, that if $W=T/2^kP$,

$$\mu'_{s-1}(W) = 2^{-k}P^{-1} \mu'_{s-1}(T) = \prod_{i=1}^k \Gamma(\frac{n-2-i}{2}+s) / \prod_{i=1}^k \Gamma(\frac{n-i}{2}).$$

Accordingly

$$\begin{aligned} f(W) &= \frac{1}{2\pi i \prod_{i=1}^k \Gamma(\frac{n-i}{2})} \int_{-c-i\infty}^{-c+i\infty} \prod_{i=1}^k \Gamma(\frac{n-2-i}{2}+s) W^{-s} ds \\ &= \frac{1}{2\pi i \prod_{i=1}^k \Gamma(\frac{n-i}{2})} \int_{-c-i\infty}^{-c+i\infty} \prod_{i=1}^k \Gamma(\frac{n-2-i}{2}-s) W^s ds \\ & (= \frac{1}{\prod_{i=1}^k \Gamma(\frac{n-i}{2})} G_{ok}^{ko}(W | \frac{1}{2}(n-3), \frac{1}{2}(n-4), \dots, \frac{1}{2}(n-k-2)) \end{aligned}$$

when written as a G function (ϕ here is the nullset)).

$$f(V) = \frac{(\frac{1}{2}n)^k}{P \prod_{i=1}^k \Gamma(\frac{n-i}{2})} G_{ok}^{ko}((\frac{1}{2}n)^k V/P | \frac{1}{2}(n-3), \frac{1}{2}(n-4), \dots, \frac{1}{2}(n-k-2)).$$

As an example of how this expression may be simplified we consider the case $k=3$. Let $(V/P) (n/2)^3 = w$, then

$$\begin{aligned} f(w) &= \frac{1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} G_{03}^{30}(w | \frac{n-3}{2}, \frac{n-4}{2}, \frac{n-5}{2}) \\ &= \frac{1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \frac{1}{2\pi i} \int_L \Gamma(\frac{n-3}{2}-s) \Gamma(\frac{n-4}{2}-2) \Gamma(\frac{n-5}{2}-2) w^s ds \end{aligned}$$

(see H.T.F.1,5.3(1), and take L as in 5.3 (3))

$$= \frac{-1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})}.$$

sum of residues of integrand in the single poles $s = \frac{n-5}{2}$ and $\frac{n-4}{2}, \frac{n-2}{2}, \frac{n}{2}$ etc.,

and in the double poles $s = \frac{n-3}{2}, \frac{n-1}{2}, \frac{n+1}{2}$, etc.

$$\begin{aligned}
 &= \frac{-1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \left\{ \lim_{s \rightarrow \frac{1}{2}} (s - \frac{n-5}{2}) \text{ integrand} + \right. \\
 &\quad \left[\lim_{s \rightarrow \frac{1}{2}} (n-4) (s - \frac{n-4}{2}) \text{ integrand} + \lim_{s \rightarrow \frac{1}{2}} (n-2) (s - \frac{n-2}{2}) \text{ integrand} + \dots \right] \\
 &\quad + \left[\lim_{s \rightarrow \frac{1}{2}} (n-3) \frac{d}{ds} (s - \frac{n-3}{2})^2 \text{ integrand} + \lim_{s \rightarrow \frac{1}{2}} (n-1) \frac{d}{ds} (s - \frac{n-1}{2})^2 \text{ integrand} \right. \\
 &\quad \left. + \dots \right] \} \\
 &= \frac{-1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \left\{ -\lim_{s \rightarrow \frac{1}{2}} \Gamma(\frac{n-3}{2} - s) \Gamma(\frac{n-4}{2} - s) \Gamma(\frac{n-3}{2} - s) w \right. \\
 &\quad - \left[\lim_{s \rightarrow \frac{1}{2}} (n-4) \Gamma(\frac{n-3}{2} - s) \Gamma(\frac{n-2}{2} - s) \Gamma(\frac{n-5}{2} - s) w^s + \dots \right] \\
 &\quad + \left[\lim_{s \rightarrow \frac{1}{2}} (n-3) \frac{d}{ds} \Gamma(\frac{n-1}{2} - s) \Gamma(\frac{n-4}{2} - s) \Gamma(\frac{n-1}{2} - s) w^s + \dots \right] \} \\
 &\quad \frac{\frac{n-5}{2} - s}{\frac{n-5}{2} - s} \\
 &= \frac{-1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \left\{ -\Gamma(1) \Gamma(\frac{1}{2}) \Gamma(1) w^{\frac{1}{2}(n-5)} \right. \\
 &\quad - [\Gamma(\frac{1}{2}) \Gamma(1) \Gamma(-\frac{1}{2}) w^{\frac{1}{2}(n-4)} + \Gamma(-\frac{1}{2}) \Gamma(\frac{1}{2}) \Gamma(-1\frac{1}{2}) w^{\frac{1}{2}(n-2)} \\
 &\quad + \Gamma(-1\frac{1}{2}) \Gamma(\frac{1}{2}) \Gamma(-2\frac{1}{2}) w^{\frac{1}{2}n} + \dots] \\
 &\quad + \left[\lim_{s \rightarrow \frac{1}{2}} (n-3) < -\psi(\frac{n-4}{2} - s) - \psi(\frac{n-1}{2} - s) + \frac{1}{\frac{n-5}{2} - s} + 1 \text{ nw} > \right. \\
 &\quad \left. \Gamma(\frac{n-1}{2} - s) \Gamma(\frac{n-4}{2} - s) \Gamma(\frac{n-1}{2} - s) w^s + \dots \right] \}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \{ \sqrt{\pi} w^{\frac{1}{2}(n-5)} - \pi w^{\frac{1}{2}(n-4)} \\
 &\quad [\frac{2}{1} - \frac{2}{1} \cdot \frac{2.2}{1.3} \frac{w}{1!} + \frac{2.2}{1.3} \cdot \frac{2.2.2}{1.3.5} \frac{w^2}{2!} - \dots] \\
 &\quad - [(-\psi(1) - \psi(-\frac{1}{2}) - \psi(1) - \frac{1}{1} + \ln w) \Gamma(1) \Gamma(-\frac{1}{2}) \Gamma(\frac{1}{-1}) w^{\frac{1}{2}(n-3)} + \\
 &\quad (-\psi(1) - \psi(-1\frac{1}{2}) - \psi(1) - \frac{1}{1} + \frac{(-2)}{(-2)} \frac{(-1)}{(-1)} + \ln w) \\
 &\quad \Gamma(1) \Gamma(-1\frac{1}{2}) \Gamma(1) \frac{1}{-1} \frac{1}{(-2)(-1)} w^{\frac{1}{2}(n-1)} + \dots] \} \\
 &= \frac{1}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \{ \sqrt{\pi} w^{\frac{1}{2}(n-5)} - \pi w^{\frac{1}{2}(n-4)} \\
 &\quad [\frac{2}{1} + \sum_{r=1}^{\infty} \frac{(-2^4 w)^r (r-1)!}{(2r-1)! (2r+1)!}] + \sum_{r=1}^{\infty} [-3\gamma - 2\ln 2 + 2 \sum_{m=1}^{2r-1} \frac{1}{m} + \sum_{m=1}^r \frac{1}{m} - \ln w] \\
 &\quad \frac{2^{2r-1} \sqrt{\pi} w^{\frac{1}{2}(n-5) + r}}{(2r-1)! r!} \} \text{ where } \psi(z) = \Gamma'(z)/\Gamma(z) \text{ and } \gamma = \text{Euler's constant}
 \end{aligned}$$

= 0.5772156649 ... (and we use H.T.F.1,1.7(8) and (12))

$$\begin{aligned}
 \text{So } f(V) &= \frac{\sqrt{\pi} V^{\frac{1}{2}(n-5)} (n^3/2^3 P)^{\frac{1}{2}(n-3)}}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})} \{ 1 - \sqrt{\pi} (n^3 V/2^3 P)^{\frac{1}{2}} [\frac{2}{1} + \\
 &\quad \sum_{r=1}^{\infty} \frac{(-2n^3 V/P)^r (r-1)!}{(2r-1)! (2r+1)!}] + \sum_{r=1}^{\infty} [-3\gamma - 2\ln 2 + \sum_{m=1}^{2r-1} \frac{1}{m} + \sum_{m=1}^r \frac{1}{m} - \ln \frac{n^3 V}{2^3 P}] \\
 &\quad \frac{2^{2r-1} (n^3 V/2^3 P)^r}{(2r-1)! r!} \} .
 \end{aligned}$$

In the cases $k=1, 2$ and 4 we have fairly simple results: Expanding $-\frac{1}{2}\text{tr}\underline{P}^{-1}\underline{T}$ in $f(\underline{T})$ (section 3.1) in terms of the cofactors of $\underline{P}$ and $\underline{T}$ (similar to the expansion in section 2), integrating out the non-diagonal elements of $\underline{T}$ as conditional normal variates, and then

transforming from $T, T_A, T_{AB}, \text{ etc.}$, to say $W=V/P, W_1=V_x^2/P\rho_{11}, W_2=V_{xy}^2/P\rho_{11}\rho_{22}, \text{ etc.}$, we find that

$$\text{if } k=1, \quad f(W) = \frac{(\frac{1}{2}n)^{\frac{1}{2}(n-1)}}{\Gamma(\frac{n-1}{2})} e^{-\frac{1}{2}nW} W^{\frac{1}{2}(n-3)},$$

$$\text{if } k=2, \quad f(W, W_1) = \frac{(\frac{1}{2}n)^{\frac{1}{2}(n-1) + \frac{1}{2}(n-2)}}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2})} e^{-\frac{1}{2}nW/W_1 - \frac{1}{2}nW_1} W^{\frac{1}{2}(n-4)} W_1^{-\frac{1}{2}},$$

$$\text{if } k=3, \quad f(W, W_1, W_2) = \frac{(\frac{1}{2}n)^{\frac{1}{2}(n-1) + \frac{1}{2}(n-2) + \frac{1}{2}(n-3)}}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2})}$$

$$e^{-\frac{1}{2}nW/W_1 - \frac{1}{2}nW_1/W_2 - \frac{1}{2}nW_2/W_3} W^{\frac{1}{2}(n-5)} W_1^{-\frac{1}{2}} W_2^{-\frac{1}{2}},$$

$$\text{if } k=4, \quad f(W, W_1, W_2, W_3) = \frac{(\frac{1}{2}n)^{\frac{1}{2}(n-1) + \frac{1}{2}(n-2) + \frac{1}{2}(n-3) + \frac{1}{2}(n-4)}}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2}) \Gamma(\frac{n-3}{2}) \Gamma(\frac{n-4}{2})}$$

$$e^{-\frac{1}{2}nW/W_1 - \frac{1}{2}nW_1/W_2 - \frac{1}{2}nW_2/W_3 - \frac{1}{2}nW_3/W_4}$$

$$W^{\frac{1}{2}(n-6)} W_1^{-\frac{1}{2}} W_2^{-\frac{1}{2}} W_3^{-\frac{1}{2}},$$

etc.

Integrate out $W_1, W_3, W_5, \text{ etc.}$, as conditional kappa variates. Apply $K_{\frac{1}{2}}(w) = \sqrt{\pi/2w} e^{-w}$ and once again the kappa integral and also the duplication formula for the gamma function, then one finds that

$$\text{if } k=2, \quad f(W) = \frac{(\frac{1}{2}n)^{\frac{1}{2}(n-1) + \frac{1}{2}(n-2)}}{\Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2})} 2 W^{1/4} K_{\frac{1}{2}}(nW^{\frac{1}{2}}) W^{\frac{1}{2}(n-4)}$$

$$= \frac{n^{n-2}}{2 \Gamma(n-2)} e^{-n\sqrt{W}} W^{\frac{1}{2}(n-4)}$$

$$\text{and if } k=4, \quad f(W) = \frac{n^{2n-6}}{\Gamma(n-2) \Gamma(n-4)} K_2(2nW^{1/4}) W^{1/2(n-5)}.$$

We finally have the following:

$$k=1: f(V) = \frac{n^{1/2(n-1)}}{(2P)^{1/2(n-1)} \Gamma(\frac{n-1}{2})} e^{-nV/2P} V^{1/2(n-3)};$$

$$k=2: f(V) = \frac{n^{n-2}}{2P^{1/2(n-2)} \Gamma(n-2)} e^{-n\sqrt{V/P}} V^{1/2(n-4)};$$

$$k=4: f(V) = \frac{n^{2n-6}}{P^{1/2(n-3)} \Gamma(n-2) \Gamma(n-4)} K_2(2n\sqrt[4]{V/P}) V^{1/2(n-5)}.$$

The method followed above appears to be new, and so is also the last result. Tables of the kappa function exist.

6. The distributions of certain products in terms of the G function

6.1 : Products of gamma and normal variates

From the previous section we see that $W/W_1, W_1/W_2, W_2/W_3, \dots, W_{k-1}/W_k$ are independent gamma variates, and that W is their product; this is known. The product of any number of independent gamma or normal variates may therefore also be expressed in terms of the G function and then computed by an expansion as that given for $f(V)$ in the case $k=3$. E.g. if x, y and z are independent *gamma* variates with respective order parameters m, n and p , but the same scale parameter 1, then

$$f(x,y,z) = \frac{1}{\Gamma(m) \Gamma(n) \Gamma(p)} e^{-(x+y+z)} x^{m-1} y^{n-1} z^{p-1},$$

and

$$f(u,v,w) = \frac{1}{\Gamma(m) \Gamma(n) \Gamma(p)} e^{-u/v-v/w-w} u^{m-1} v^{n-m-1} w^{p-n-1},$$

and

$$f(u) = \frac{1}{\Gamma(m) \Gamma(n) \Gamma(p)} G_{03}^{30}(u |_{m-1, n-1, p-1}, \phi,$$

as we see if we compare the expression for $f(W, W_1, W_2)$ with that for $f(V)$ in terms of the G function.

If x , y and z are independent *normal* $(0,1)$ variates, then $\frac{1}{2}x^2$, $\frac{1}{2}y^2$ and $\frac{1}{2}z^2$ are independent gamma variates with order parameters each $\frac{1}{2}$ and all three scale parameters 1, so $(xyz)^2/8$ is distributed as u above with $m=n=p=\frac{1}{2}$, and the product xyz is therefore distributed as follows:

$$f(xyz) = \frac{xyz}{2^{3-1}\Gamma^{3(\frac{1}{2})}} G_{03}^{30} \left(\frac{(xyz)^2}{2^3} \middle| -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2} \right)$$

Different scale parameters actually offer no problem (different variances hence also do not), non-centrality with regard to means does offer a problem though: If x , y and z have scale parameters r , s and t , then rx , sy and tz have scale parameter 1.

6.2 : Products of beta variates of the second kind (Be_2 variates)

This matter will be considered at the end of the next section.

7. The distribution of the ratio of two variance determinants

Let V_1 and V_2 be independent variance determinants respectively of orders h and k , sample sizes m and n , and population variance determinants P_1 and P_2 , then by H.T.F.1, 5.3(8), we find in the usual manner that unbiased estimators $\hat{P}_1$ and $\hat{P}_2$ for P_1 and P_2 are

$$V_1 \prod_{i=1}^h \frac{m}{m-i} \text{ and } V_2 \prod_{j=1}^k \frac{n}{n-j} \text{ respectively.}$$

If $F' = (\hat{P}_1/P_1)/(\hat{P}_2/P_2)$, it follows that

$$f(F') = \left(\left(\frac{1}{2}m \right)^h \prod_{j=1}^k \frac{n}{n-j} / \left(\frac{1}{2}n \right)^k \prod_{i=1}^h \left(\frac{m-i}{2} \right) \prod_{j=1}^k \left(\frac{n-j}{2} \right) \right)^{-1} \\ G_{kh}^{hk} \left(\frac{\left(\frac{m}{2} \right)^h \prod_{i=1}^h \frac{m-i}{2}}{\left(\frac{n}{2} \right)^k \prod_{j=1}^k \frac{n-j}{2}} F' \middle| \begin{matrix} -\frac{1}{2}(m-3), -\frac{1}{2}(m-4), \dots, -\frac{1}{2}(m-h-2) \\ \frac{1}{2}(n-1), \frac{1}{2}(n-2), \dots, \frac{1}{2}(n-k) \end{matrix} \right)$$

by T.I.T.2, 20.5(14).

The case $h=k=3$ which we treat now, shows the general method for obtaining $f(F')$ for arbitrary natural h and k .

$$f(F') = \prod_{i=1}^3 \frac{(m-i)/(n-i)}{\Gamma(\frac{m-i}{2}) \Gamma(\frac{n-i}{2})} G_{33}^{33} \left(F' \prod_{i=1}^3 \frac{m-i}{n-i} \middle| \begin{matrix} -\frac{1}{2}(m-3) -\frac{1}{2}(m-4) -\frac{1}{2}(m-5) \\ \frac{1}{2}(n-1) \quad \frac{1}{2}(n-2) \quad \frac{1}{2}(n-3) \end{matrix} \right)$$

$$= -K \times \text{the sum of the residues of } \Gamma\left(\frac{n-1}{2} - s\right) \Gamma\left(\frac{n-2}{2} - s\right) \Gamma\left(\frac{n-3}{2} - s\right)$$

$$\Gamma\left(\frac{m-1}{2} + s\right) \Gamma\left(\frac{m-2}{2} + s\right) \Gamma\left(\frac{m-3}{2} + s\right) w^s \text{ in the single poles } s = \frac{n-3}{2} \text{ and}$$

$$\frac{n-2}{2}, \frac{n}{2}, \frac{n+2}{2}, \text{ etc., and in the double poles } s = \frac{n-1}{2}, \frac{n+1}{2}, \frac{n+3}{2}, \text{ etc.}$$

(where K is the norming constant of the p.d.f., and w the argument of the G function, above).

$$\begin{aligned} &= K \sqrt{\pi} w^{\frac{1}{2}(n-3)} \left\{ \Gamma\left(\frac{m+n-4}{2}\right) \Gamma\left(\frac{m+n-5}{2}\right) \Gamma\left(\frac{m+n-6}{2}\right) \right. \\ &\quad - \sqrt{\pi} w^{\frac{1}{2}} \left[\frac{2}{1} \Gamma\left(\frac{m+n-3}{2}\right) \Gamma\left(\frac{m+n-4}{2}\right) \Gamma\left(\frac{m+n-5}{2}\right) + \right. \\ &\quad \left. \sum_{r=1}^{\infty} \frac{(-2^4 w)^r (r-1)!}{(2r-1)! (2r+1)!} \Gamma\left(\frac{m+n+2r-3}{2}\right) \right. \\ &\quad \left. \Gamma\left(\frac{m+n+2r-4}{2}\right) \Gamma\left(\frac{m+n+2r-5}{2}\right) \right] , \\ &\quad + \sum_{r=1}^{\infty} \left[-3\gamma - 2 \ln w + 2 \sum_{m=1}^{2r-1} \frac{1}{m} + \sum_{m=1}^r \frac{1}{m} - \ln w \right. \\ &\quad \left. - \psi\left(\frac{m+n+2r-4}{2}\right) - \psi\left(\frac{m+n+2r-5}{2}\right) - \psi\left(\frac{m+n+2r-6}{2}\right) \right] \frac{2^{2r-1} w^r}{(2r-1)! r!} \\ &\quad \left. \Gamma\left(\frac{m+n+2r-4}{2}\right) \Gamma\left(\frac{m+n+2r-5}{2}\right) \Gamma\left(\frac{m+n+2r-6}{2}\right) \right\} \end{aligned}$$

which we write down without difficulty from the expression for $f(w)$ just before that for $f(V)$, $k=3$, in section 5.

We also give the p.d.f.'s of $F=(\hat{P}_1/P_1)^{1/h} / (\hat{P}_2/P_2)^{1/k}$ (rather than f') in a few simple instances:

$$h=k=1: F = \frac{m}{m-1} \frac{V_1}{P_1} / \frac{n}{n-1} \frac{V_2}{P_2} \quad \text{and,}$$

$$\begin{aligned}
 f(F) &= \frac{\left(\frac{m-1}{n-1}\right)^{\frac{1}{2}(m-1)}}{B\left(\frac{m-1}{2}, \frac{n-1}{2}\right)} \frac{F^{\frac{1}{2}(m-3)}}{\left(\frac{m-1}{n-1} F + 1\right)^{\frac{1}{2}(m-1) + \frac{1}{2}(n-1)}} \text{ (known)} \\
 &= 2 \left(\frac{m-1}{n-1}\right)^{\frac{1}{2}(m+1)} \frac{\Gamma\left(\frac{m+n+2}{4}\right) \Gamma\left(\frac{m+n}{4}\right) \Gamma\left(\frac{m+n-2}{4}\right)}{\Gamma\left(\frac{m+1}{4}\right) \Gamma\left(\frac{m-1}{4}\right) \Gamma\left(\frac{n+1}{4}\right) \Gamma\left(\frac{n-1}{4}\right)} F^{\frac{1}{2}(m-1)} \\
 &\quad {}_2F_1\left(\frac{m+n+2}{4}, \frac{m+n}{4}, \frac{m+n}{2}; 1 - \left(\frac{m-1}{n-1}\right)^2 F^2\right).
 \end{aligned}$$

$$h=k=2: F = \sqrt{\frac{m^2}{(m-1)(m-2)} \frac{V_1}{P_1}} / \sqrt{\frac{n^2}{(n-1)(n-2)} \frac{V_2}{P_2}} \text{ and}$$

$$\begin{aligned}
 f(F) &= \frac{\left(\frac{m-1}{n-1} \frac{m-2}{n-2}\right)^{\frac{1}{2}(2m-2)}}{B(m-2, n-2)} \frac{F^{m-3}}{\left(\sqrt{\frac{(m-1)(m-2)}{(n-1)(n-2)}} F + 1\right)^{(m-2) + (n-2)}} \\
 &= 2 \left(\frac{m-1}{n-1} \frac{m-2}{n-2}\right)^{\frac{1}{2}(m-1)} \frac{\Gamma\left(\frac{2m+2n-4}{4}\right) \Gamma\left(\frac{2m+2n-6}{4}\right) \Gamma\left(\frac{2m+2n-8}{4}\right)}{\Gamma\left(\frac{2m}{4}\right) \Gamma\left(\frac{2m-2}{4}\right) \Gamma\left(\frac{2n}{4}\right) \Gamma\left(\frac{2n-2}{4}\right)} \\
 &\quad F^{\frac{1}{2}(2m-4)} {}_2F_1\left(\frac{2m+2n-4}{4}, \frac{2m+2n-6}{4}, \frac{2m+2n-6}{2}; 1 - \sqrt{\frac{(m-1)(m-2)}{(n-1)(n-2)}}^2 F^2\right).
 \end{aligned}$$

$$h=k=4: F = \sqrt[4]{\frac{m^4}{(m-1)(m-2)(m-3)(m-4)} \frac{V_1}{P_1}} / \sqrt[4]{\frac{n^4}{(n-1)(n-2)(n-3)(n-4)} \frac{V_2}{P_2}} \text{ and}$$

$$\begin{aligned}
 f(F) &= 2 \left(\frac{m-1}{n-1} \frac{m-2}{n-2} \frac{m-3}{n-3} \frac{m-4}{n-4}\right)^{\frac{1}{8}(4m-8)} \\
 &\quad \frac{\Gamma\left(\frac{4m+4n-16}{4}\right) \Gamma\left(\frac{4m+4n-24}{4}\right) \Gamma\left(\frac{4m+4n-32}{4}\right)}{\Gamma\left(\frac{4m-8}{4}\right) \Gamma\left(\frac{4m-16}{4}\right) \Gamma\left(\frac{4n-8}{4}\right) \Gamma\left(\frac{4n-16}{4}\right)}
 \end{aligned}$$

$$F^{\frac{1}{2}(4m-10)} {}_2F_1\left(\frac{4m+4n-16}{4}, \frac{4m+4n-24}{4}; \frac{4m+4n-24}{4};\right.$$

$$\left. 1 - \sqrt[4]{\frac{(m-1)(m-2)(m-3)(m-4)}{(n-1)(n-2)(n-3)(n-4)}}\right).$$

$$h=1, k=2: F = \frac{m}{(m-1)} \frac{V_1}{P_1} / \sqrt{\frac{n^2}{(n-1)(n-2)} \frac{V_2}{P_2}} \text{ and}$$

$$f(F) = \frac{1}{2^{\frac{1}{2}(m-2)}} \Gamma(m+2n-1) \Gamma$$

$$f(F) = \frac{1}{2^{\frac{1}{2}(m-2)}} \frac{\Gamma(\frac{m+2n-1}{4}) \Gamma(\frac{m+2n-3}{4}) \Gamma(\frac{m+2n-5}{4})}{\Gamma(\frac{m+1}{4}) \Gamma(\frac{m-1}{4}) \Gamma(\frac{n-1}{2}) \Gamma(\frac{n-2}{2})}$$

$$\left(\frac{m-1}{\sqrt{(n-1)(n-2)}}\right)^{\frac{1}{2}(m+1)} F^{\frac{1}{2}(m-1)}$$

$${}_2F_1\left(\frac{m+2n-1}{4}, \frac{m+2n-3}{4}, \frac{m+2n-3}{2}; 1 - \frac{1}{4} \left(\frac{m-1}{\sqrt{(n-1)(n-2)}}\right)^2 F^2\right).$$

$$h=1, k=4: f(F) = \frac{1}{2^m} \frac{\Gamma(\frac{m+2n-7}{4}) \Gamma(\frac{m+4n-15}{4}) \Gamma(\frac{m+4n-9}{4}) \Gamma(\frac{m+4n-17}{4})}{\Gamma(\frac{m+1}{4}) \Gamma(\frac{m-1}{4}) \Gamma(n-2) \Gamma(n-4) \Gamma(\frac{m+n-12}{2})}$$

$$\left(\frac{m-1}{\sqrt{(n-1)(4)}}\right)^{\frac{1}{2}(m+1)}$$

$$F^{\frac{1}{2}(m-1)} {}_2F_1\left(\frac{m+4n-7}{4}, \frac{m+4n-15}{4}, \frac{m+4n-12}{2};\right.$$

$$\left. 1 - \frac{1}{16} \left(\frac{m-1}{\sqrt{(n-1)(4)}}\right)^2 F^2\right).$$

$$h=2, k=4: f(F) = \frac{1}{2^{m-2}} \frac{\Gamma(\frac{2m+4n-10}{4})\Gamma(\frac{2m+4n-18}{4})\Gamma(\frac{2m+4n-12}{4})\Gamma(\frac{2m+4n-20}{4})}{\Gamma(\frac{m-1}{2})\Gamma(\frac{m-2}{2})\Gamma(n-2)\Gamma(n-4)\Gamma(\frac{2m+4n-15}{2})} F^{m-2}$$

$${}_2F_1\left(\frac{2m+4n-10}{4}, \frac{2m+4n-18}{4}; \frac{2m+4n-15}{2};\right.$$

$$\left.1 - \left(\frac{\sqrt{(m-1)(2)}}{2\sqrt[4]{(n-1)(4)}}\right)^2 F^2\right) \left(\frac{\sqrt{(m-1)(2)}}{4\sqrt[4]{(n-1)(4)}}\right)^{m-1}$$

We finally turn to the distribution of the product (or quotient) of independent beta variates of the second kind: If x and y are independent $\text{Be}_2(m, n)$ and $\text{Be}_2(p, q)$ variates, and if u denotes their product, then

$$f(u) = \frac{B(m+q, n+p)}{B(m, n)B(p, q)} {}_2F_1(m+n, m+q; m+n+p+q; 1-u) u^{m-1}$$

(by T.I.T.1, 6.2(23), and H.T.F.1, 2.9(8)), and we may, simultaneously, exchange m and p , n and q . If the quotient x/y is denoted by v , then

$$f(v) = \frac{B(m+p, n+q)}{B(m, n)B(p, q)} {}_2F_1(m+n, m+p; m+n+p+q; 1-v) v^{m-1},$$

for if y is a $\text{Be}_2(p, q)$ variate then $1/y$ is a $\text{Be}_2(q, p)$ variate, and $x/y = x \cdot 1/y$.

If $m=p+\frac{1}{2}$ and $n=q+\frac{1}{2}$, and $z=\sqrt{xy}=\sqrt{u}$, we find that

$$f(z) = 2 \frac{B(p+q+\frac{1}{2}, p+q+\frac{1}{2})}{B(p+\frac{1}{2}, q+\frac{1}{2})B(p, q)} F_1(p+q+1, p+q+\frac{1}{2}; 2p+2q+1; 1-z^2) z^{2p};$$

however, if we begin with two independent gamma variates, and work with the gamma p.d.f.'s in both the exponential form and re-written by means of $e^{-x} = (2x/\pi)^{1/2} K_{1/2}(x)$, then we find by T.I.T.1, 6.8(47), that

$${}_2F_1(a+1, a+\frac{1}{2}; 2a+1; 1-x^2) = 2^{2a}/x(1+x)^{2a},$$

and if we substitute this in the expression for $f(z)$ we find that

$$f(z) = \frac{1}{B(2p, 2q)} z^{2p-1}/(1+z)^{2p+2q},$$

i.e. if x and y are $\text{Be}_2(p+\frac{1}{2}, q+\frac{1}{2})$ and $\text{Be}_2(p, q)$ variates, then $\sqrt{xy}$ is a $\text{Be}_2(2p, 2q)$ variate.

Now, the preceding is a special case. Suppose therefore that we have k independent variates $z_1, z_2, \dots, z_k$ that are respectively $\text{Be}_2(m_j, n_j)$ variates, $j=1, \dots, k$. We may regard each z_j as the quotient of two independent gamma variates x_j and y_j with order parameters m_j and n_j (and scale parameters both 1, for convenience sake), in other words the product $z_1 z_2 \dots z_k$ as the quotient of two independent products of, respectively, independent gamma variates x_j and y_j . It follows that, if we write u for $z_1 z_2 \dots z_k$,

$$g \quad f(u) = \frac{1}{\prod_{j=1}^k \Gamma(m_j) \Gamma(n_j)} G_{kk}^{kk}(u | \begin{smallmatrix} -n_1, -n_2, \dots, -n_k \\ m_1-1, m_2-1, \dots, m_k-1 \end{smallmatrix}).$$

8. The distribution of t , p and q

If, as in standardised educational testing, the population mean and variance are known, $f(t, p, q)$ should lead to more satisfactory statistical decisions than $f(T, p, q)$ (or, for that matter, $f(V, \bar{x}, \bar{y})$). Intractable integration limits prevented the derivation of $f(t, p, q)$ from $f(a, b, d, p, q)$ in manageable form, and so an attempt was made to obtain it through characteristic functions.

If $k=n=2$, the c.f. of $t^{1/2}/P$, p/P and q/P is easily found, and when inverted, gives

$$f(t^{1/2}/P, p/P, q/P) = \frac{P^{2^{1/2}}}{4\pi^{1/2}} e^{-t/s - s/4P} s^{-1/2}, \quad -\infty < t^{1/2}/P < \infty,$$

where $s = p^2 - 2\rho pq + q^2$. But this expression could not be generalised. So a generalisation of the c.f. was proposed:

$$\phi(a, \beta, \gamma; t^{1/2}/P, p/P, q/P; k=2, n \geq 2) = \left(\frac{P}{P+a^2}\right)^{1/2n} e^{-n(\beta^2 + 2\rho\beta\gamma + \gamma^2)/2P(P+a^2)},$$

which gives the correct marginal c.f.'s for $t^{1/2}/P$ and $(p/P, q/P)$ (e.g. $t^{1/2}/P$ has a Laplace distribution with parameters $n-2$ and $\sqrt{P}$) and is moreover suggested by the c.f. in section 2. Inversion leads to

$$f(t^{1/2}/P, p/P, q/P) = \frac{P^{1/2}}{\sqrt{n} (2\pi)^{1/2}} e^{-s/2nP - nt/2s} s^{-1/2} \\ (1 - \frac{n}{2P} \text{He}_2(P\sqrt{nt/s}))^{-1/2(n-2)}$$

where $(\text{He}_2(x))' = \text{He}_{2r}(x) = (2r)\text{th}$ Hermite polynomial (of Statistics) in x .

As a possible alternative to $f(t, p, q)$ we suggest $f(t, w)$ where t is as above (2×2) and $w = (gp + hq)^2 / (n(g^2 a + 2ghd + h^2 b)) = \Sigma^2(gx + hy) / n \Sigma(gx + hy)^2$ where x and y are as in section 1 and g and h arbitrary real constants not both zero. w is seen to be a $\text{Be}(1/2, (n-1)/2)$ variate independent of t : $(gx + hy)/(g^2 + h^2)^{1/2}$ is an $N(0, 1)$ variate because x and y are such, so w has the same distribution as $p^2/n / (A + p^2/n)$ which is a

$\text{Be}(1/2, n-1/2)$ variate since p and a are independent and p^2/n and A are chisquared variates with respectively 1 and $n-1$ degrees of freedom; further, $(p^2/n)/(A+p^2/n)$ is independent of its denominator, and so w is independent of $g^2a+2ghd+h^2b = (g,h)\underline{t}(h)$, which suggests that w is independent of $\underline{t}$ since g and h are constants, and if we take $\underline{t}(a,b,d,p,q)$ and transform from p to w and then integrate out q , then we indeed find that $f(a,b,d,w)$ factors into $f(a,b,d)$ and $f(w)$. We may extend w to the k -variate normal distribution, e.g. in the trivariate case we have $w = \Sigma^2 (gx+hy+\ell z) / n\Sigma(gx+hy+\ell z)^2$ (x,y,z as in section 1, and g,h,ℓ arbitrary real constants not all zero), and we find that in each case w is not only independent of $\underline{t}$ but also (unlike Hotelling's T^2) a $\text{Be}(1/2, n-1/2)$ variate.

$([n-1]w/1-w)^{1/2}$ (which is equal to $\sqrt{n-1}(g\bar{x}+h\bar{y})/\sqrt{(g^2s_x^2+2rghs_xs_y+h^2s_y^2)}$ when $k=2$) reduces to $\sqrt{n-1}(x-\mu_x)/s_x$ when $g=1$ and $h=\ell=\dots=0$, so it is a generalisation of Student's ratio to the case of the k -variate normal distribution; but, unlike Hotelling's T , it is also distributed as Student's ratio (with $n-1$ degrees of freedom) for all k . When

$h=-k=1$ and the other constants zero, it is equal to

$$\sqrt{n-1}([\bar{x}-\bar{y}] - [\mu_x - \mu_y]) / \sqrt{s_x^2 - 2rs_{xy} + s_y^2}$$

In the case of the normal distributions we commenced with, μ_x, μ_y etc. were of course zero.)

The author wishes to thank his promoter, Prof. H.S. Steyn, for suggesting this investigation and for his guidance during the preparation of the thesis.

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